

BENEDIKT BUCHECKER

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PERSONAL INFORMATION

Date of Birth: 05.07.2000

Place of Birth: Vienna, Austria

Citizenship: Austrian

National Registry Number: 563.44

Languages: German, English

EDUCATION

Doctoral Programme in Science - Faculty of Science

Katholieke Universiteit Leuven | Leuven, BE

Sep 2025 – today

Project: Fine asymptotics of extremal polynomials and universality limits for Coulomb gases on curves and arcs

Master of Science - Technical Mathematics

Technische Universität Wien | Vienna, AT

Sep 2023 – Jun 2025

Graduation with honors (grade average 1.0)

Thesis-Title: Asymptotics of extremal polynomials on a Jordan curve or arc

Bachelor of Science - Technical Mathematics

Technische Universität Wien | Vienna, AT

Sep 2019 – Sep 2023

Graduation with honors (grade average 1.1)

Early Student Award from the Austrian Mathematic Society (ÖMG)

Thesis-Title: C^* -Algebras

High School

Realgymnasium GRG 16 Maroltingergasse | Vienna, AT

Sep 2015 – Jun 2018

Graduation with honors (grade average 1.0)

Award for pre-scientific thesis: Dr. Hans-Riegl Fachpreis in Mathematik

EXPERIENCE

PhD Fellowship fundamental research

Katholieke Universiteit Leuven | Leuven, BE

Nov 2025 – today

Funded by project 1154426N of the Research Foundation Flanders (FWO)

PhD Scholarship Holder

Katholieke Universiteit Leuven | Leuven, BE

Sep 2025 – Oct 2025

Funded by Odysseus Grant G0DDD23N of the Research Foundation Flanders (FWO)

Student Staff in Teaching

Technische Universität Wien | Vienna, AT

Mar 2025 – Jul 2025

Teaching Assistant for exercises in Analysis 1 for Technical Mathematics

Student Staff in Research/Administration

Technische Universität Wien | Vienna, AT

Mar 2024 – Oct 2024

Funded by project P33885-N of the Austrian Science fund FWF

Preparation of two publications

Student Staff in Teaching

Technische Universität Wien | Vienna, AT

Oct 2023 – Feb 2024

Teaching Assistant for exercises in Mathematics 1 for Civil Engineering

Exchange Semester

KTH Stockholm | Stockholm, SE

Jan 2023 – Jun 2023

Internship

Dr. Ing. h.c. F. Porsche | Stuttgart, DE

Sep 2022 – Oct 2022

Student Staff in Teaching

Technische Universität Wien | Vienna, AT

Oct 2021 – Jun 2022

Teaching Assistant for exercises in Mathematics 1&2 for Electrical Engineering and Database Systems

CONFERENCES/WORKSHOPS

OPSFA 18

Doshisha University | Kyoto, JP

Aug 2026

Poster presentation 'Chebyshev Polynomials on a Jordan Arc'

FOCM 2026

Universität Wien | Vienna, AT

Jul 2026

Poster presentation 'Chebyshev Polynomials on a Jordan Arc'

Spectral Theory by the Lakes

Lancaster University | Lancaster, UK

Apr 2026

Contributed talk 'An upper bound for the Widom factors on Jordan arcs'

Bench Math Session 2026

Jagiellonian University | Krakow, PL

Feb 2026

Contributed talk 'Extremal polynomials on a Jordan curve or arc'

Random Matrix Theory and Integrable Systems

Great Bay University | Dongguan, CN

Jan 2026

SQuaRE meeting on weighted Chebyshev/Extremal polynomials

American Institute of Mathematics | Pasadena CA, US

May 2025

Operator Theory and Indefinite Inner Products

Technische Universität Wien | Vienna, AT

Feb 2025

OTA 2024

Technische Universität Wien | Vienna, AT

Jul 2024

Contributed talk 'Extremal polynomials on a Jordan curve or arc'

OPSFOTA 2024

University of Bristol | Bristol, UK

Jul 2024

Short-Fire-Session and poster presentation 'Szegő's theorem on a Jordan arc'

ARNO 2024

Katholieke Universiteit Leuven | Leuven, BE

May 2024

Poster presentation 'Szegő's theorem on a Jordan arc'

PUBLICATIONS

Chebyshev polynomials on a Jordan arc

Preprint available: <https://arxiv.org/abs/2608.13445>

Aug 2026

Szegő's theorem on a Jordan arc

Submitted. Preprint available: <https://arxiv.org/abs/2509.12445>

Sep 2025

Asymptotics of L^r extremal polynomials for $0 < r \leq \infty$ on C^{1+} Jordan regions

Accepted. Preprint available: <https://arxiv.org/abs/2502.17616>

Feb 2025

VOLUNTARY WORK

Katholische Jungschar

Pfarre Sandeilen | Vienna, AT

Sep 2016 – Aug 2024

Weekly leisure activities for a group of children and organization of a summer camp

REFERENCES

Benjamin Eichinger

University of Lancaster | b.eichinger@lancaster.ac.uk

Aron Wennman

Katholieke Universiteit Leuven | aron.wennman@kuleuven.be

Harald Woracek

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